

Data Communication and Networking

What is a Signal?

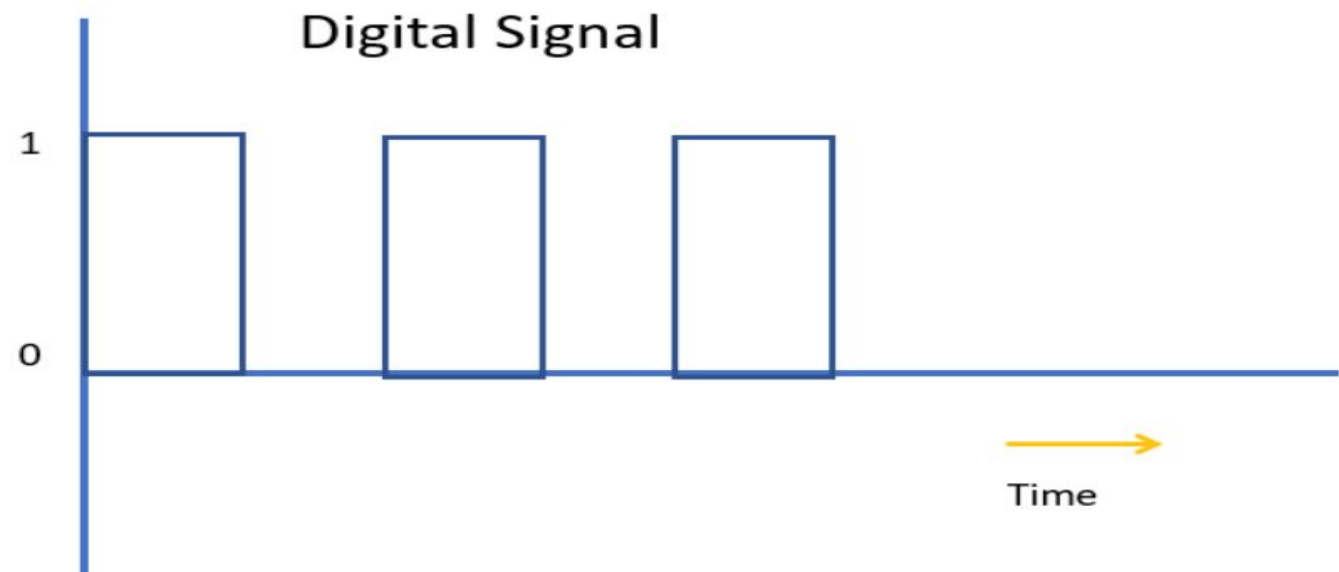
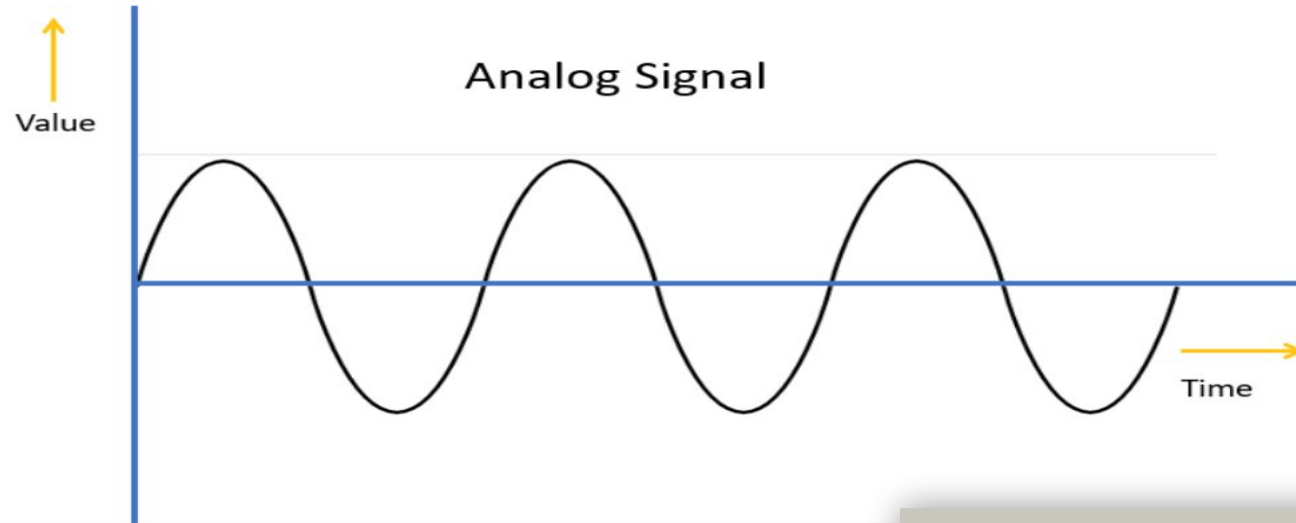
A signal is an electromagnetic or electrical current that is used for carrying data from one system or network to another.

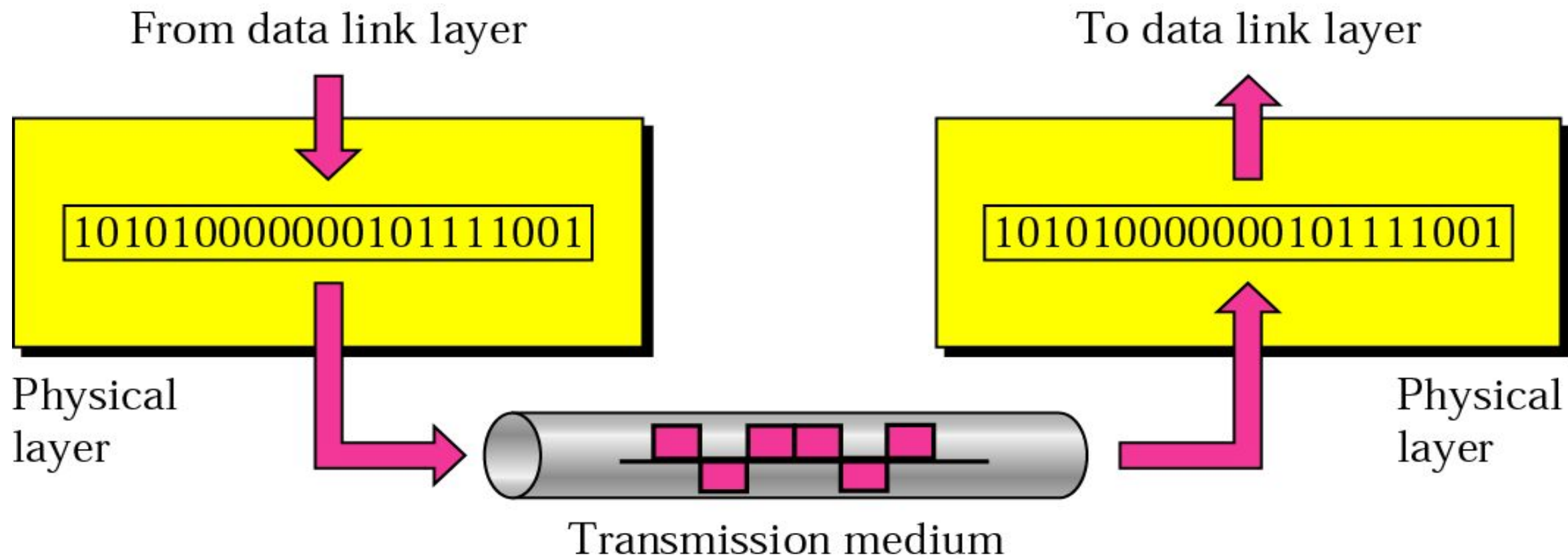
What is an Analog Signal?

An **analog signal** is a continuously varying electromagnetic wave that may be propagated over a variety of media, depending on frequency; examples are copper wire media, such as twisted pair and coaxial cable; fiber optic cable; and atmosphere or space propagation (wireless).

What is a Digital Signal?

A digital signal is a signal that is used to **represent data as a sequence of separate** values at any point in time. It can only take on one of a fixed number of values.

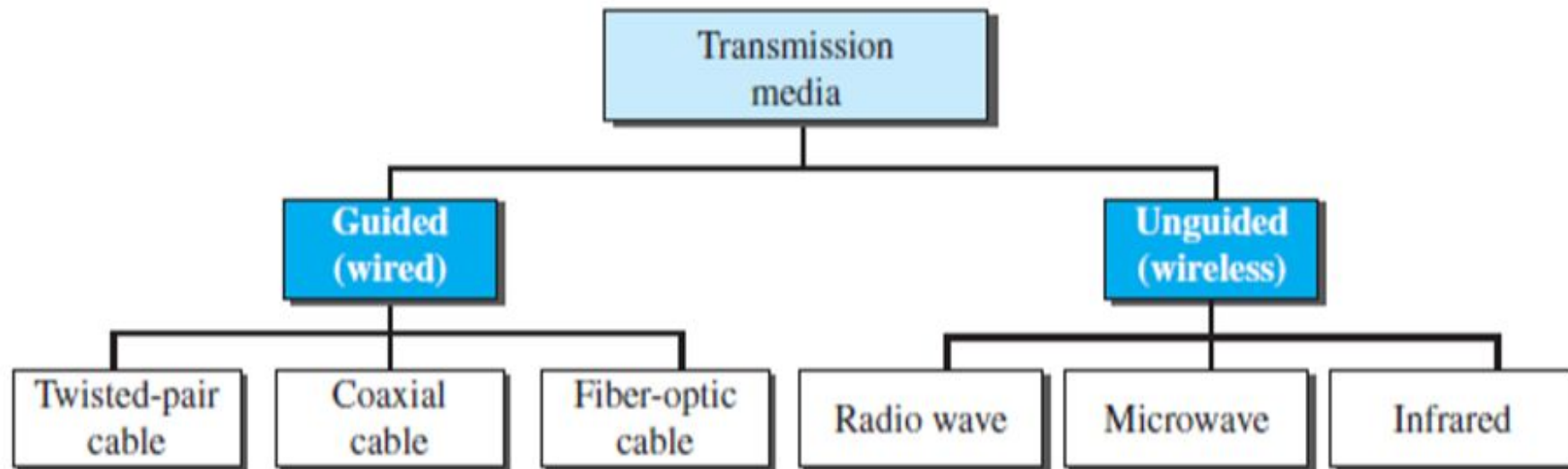




TRANSMISSION MEDIA (contd...)

In telecommunications, transmission media can be divided into two broad categories: guided and unguided.

- Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable.
- Unguided medium is free space.

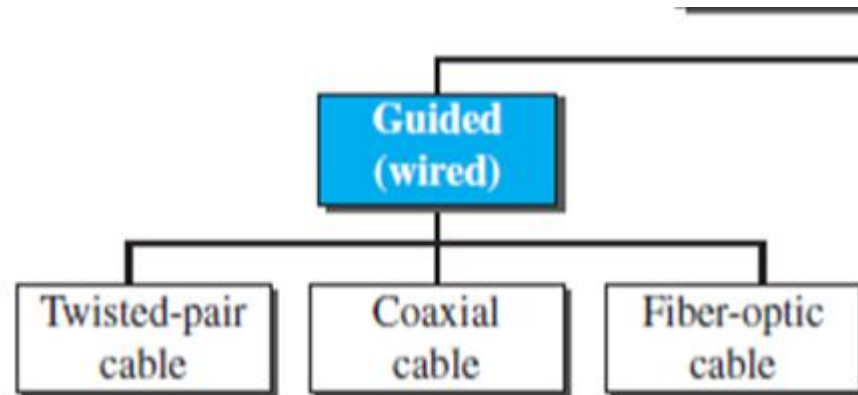


FACTORS TO BE CONSIDERED WHILE CHOOSING TRANSMISSION MEDIUM

- Transmission Rate
- Cost and Ease of Installation
- Resistance to Environmental Conditions
- Distances

GUIDED MEDIA

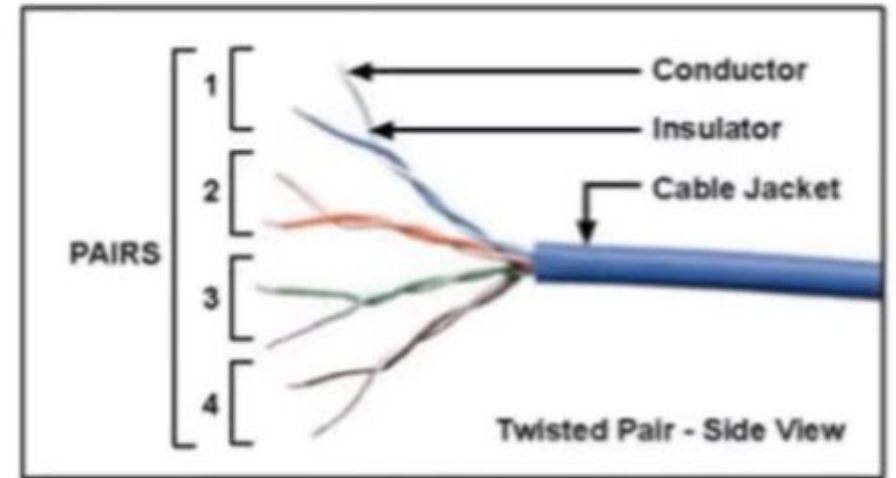
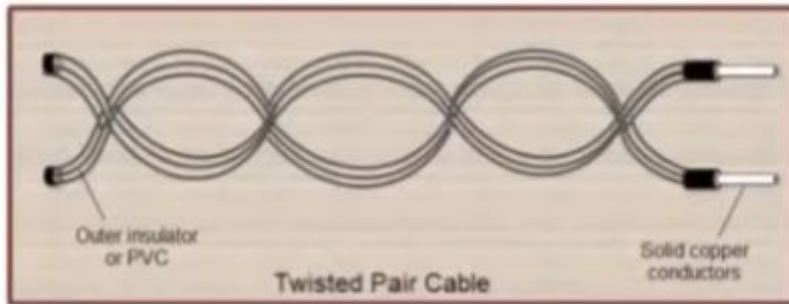
Guided media, which are those that provide a conduit from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.



TWISTED PAIR CABLE

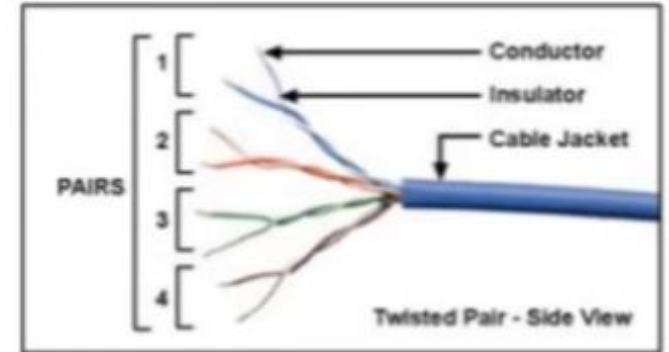
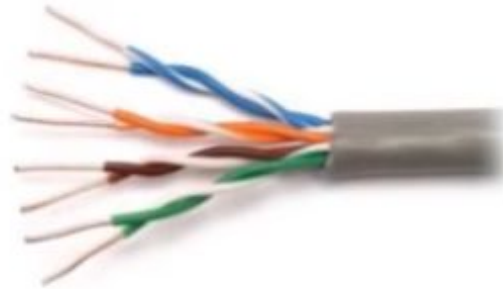
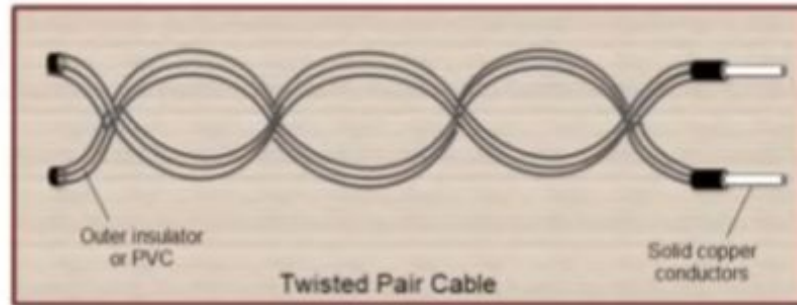
- **This cable is the most commonly used and is cheaper than others. It is lightweight, cheap, can be installed easily, and they support many different types of network.**
- **A twisted pair consists of two conductors (normally copper), each with its own plastic insulation, twisted together, as shown.**

Twisted Pair Cable



- A twisted pair cable is a pair of **copper wires**.
- Copper wires are the most common wires used for transmitting signals because of good performance at low costs.
- A twisted pair consists of **two conductors** (normally copper), each with its own **plastic insulation**, twisted together to form a single media.

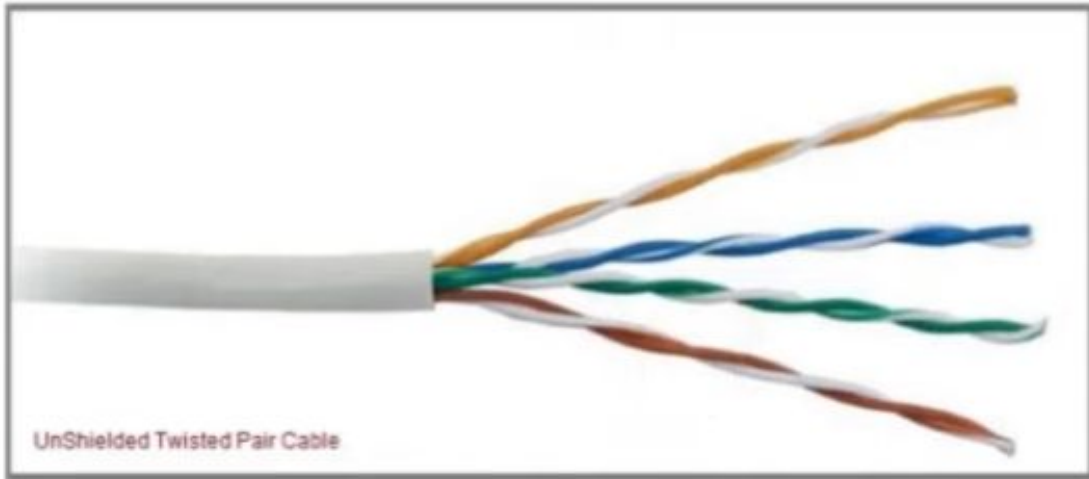
Twisted Pair Cable



- Out of these two wires, only one carries **actual signal** and another is used for ground reference.
- To identify every cable, these cables are **colour coated**.
- The twists between wires are helpful in **reducing noise** (electro-magnetic interference) and **crosstalk**.
- This type of cable is used in **telephone lines** to provide voice and data channels.
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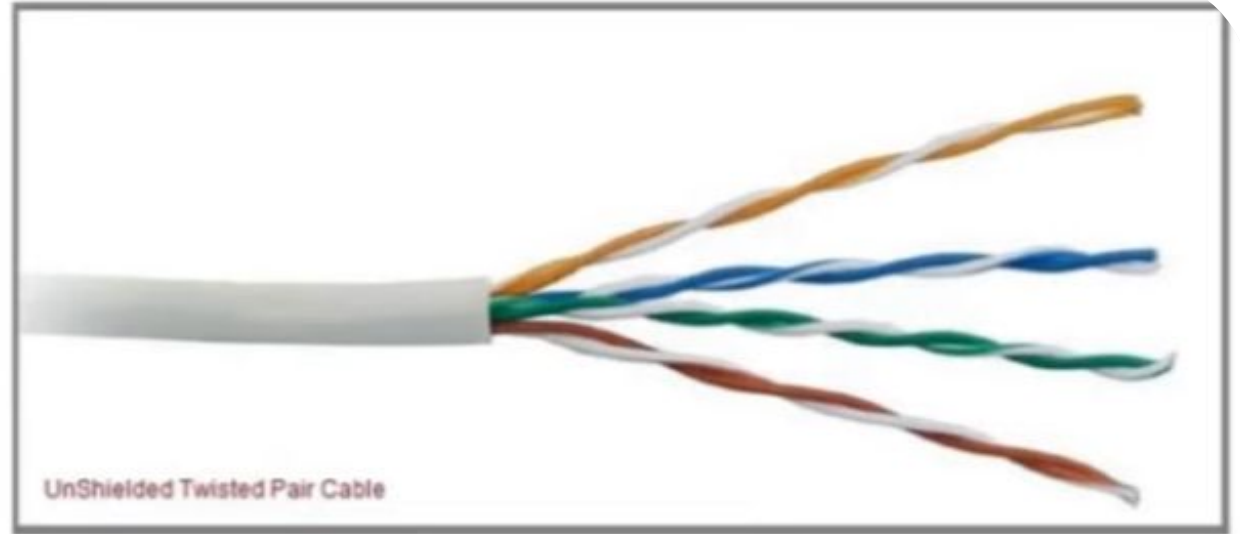
Types of Twisted Pair

- The two types of twisted pairs are:
 - 1. Unshielded twisted pair (UTP)
 - 2. Shielded twisted pair (STP)



- **Unshielded twisted pair (UTP):-**

- UTP is more common.



- UTP cost less than STP and easily available due to its many use.
- Due to its low cost, UTP cabling is used extensively for local-area networks (LANs) and telephone connections.
- UTP cables consist of 2 or 4 pairs of twisted cable.
- Cable with 2 pair use **RJ-11** connector and 4 pair cable use **RJ-45** connector.

RJ-11

6 pin



RJ-45

8 pin



- RJ-45 connectors is used with Ethernet cables in computer networking.
- RJ-11 connectors is used in connecting telephone units.
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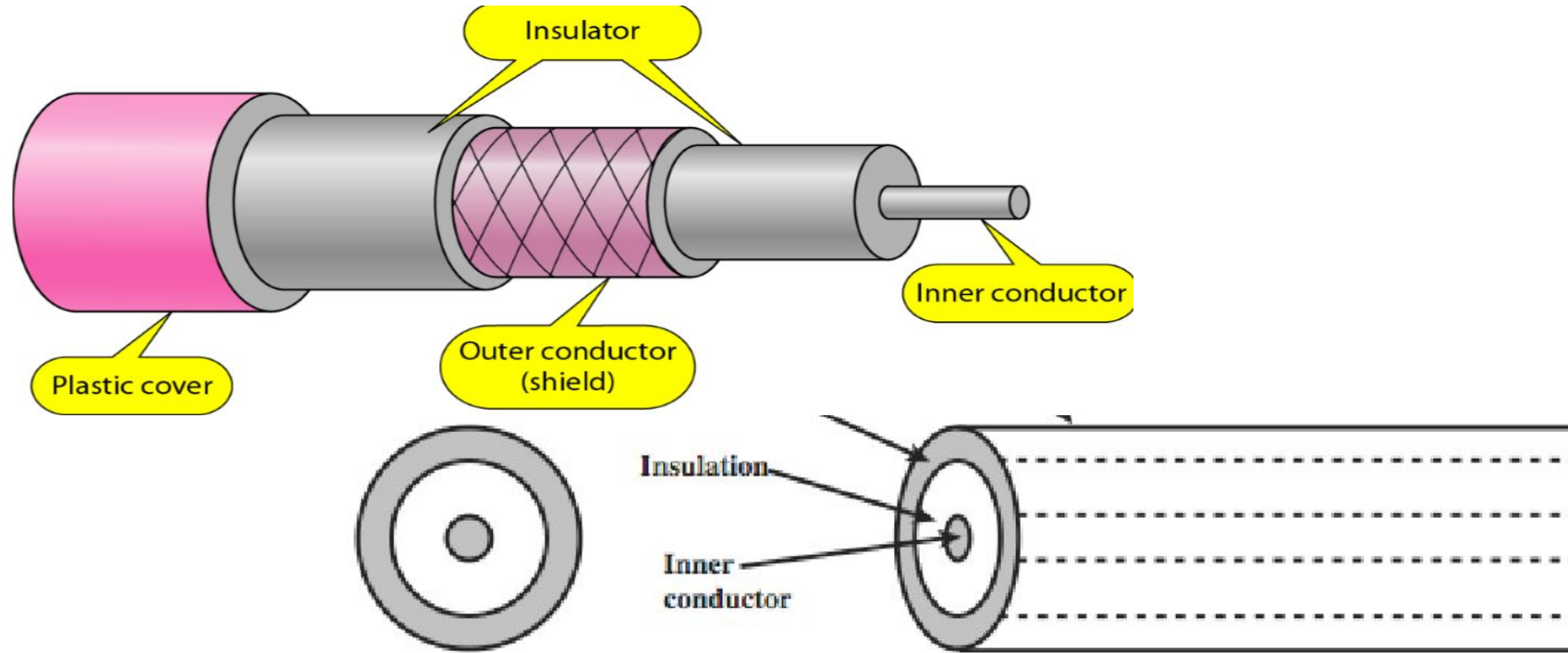
- **Shielded twisted pair (STP):-**

- This type of cable has a **metal foil covering** which encases each pair of insulator conductors.
- Electromagnetic noise penetration is prevented by metal casing. Shielding also eliminates crosstalk.
- It is similar to UTP but has a mesh shielding that's protects it from EMI which allows for higher transmission rate.

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GUIDED MEDIA – COAXIAL CABLE



- Outer conductor is braided shield
- Inner conductor is solid metal
- Separated by insulating material
- Covered by padding

TYPES OF COAXIAL CABLES

- Baseband:

Which is used for digital transmission. It is mostly used for LAN's. Baseband transmits a single signal at a time with very high speed.

- Broadband:

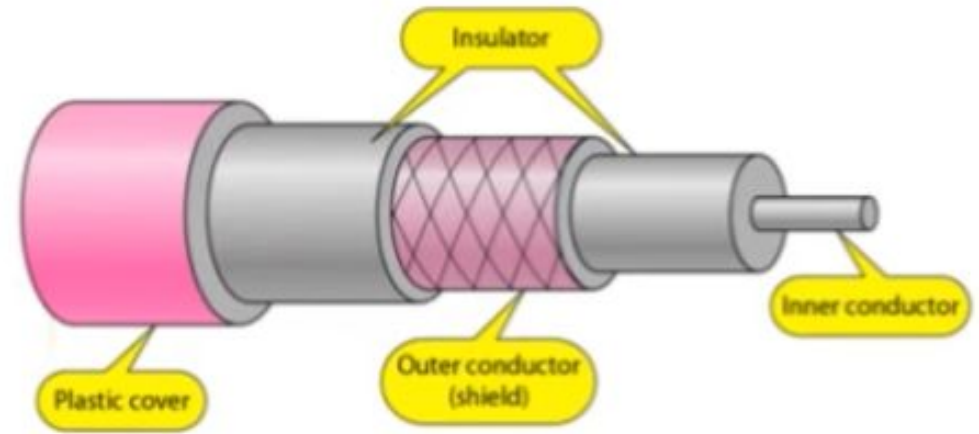
This uses analog transmission on standard cable television cabling. It transmits several simultaneous signals using different frequencies.

Coaxial Cable



- **Coaxial cables** are copper cables with better shielding than twisted pair cables, so that transmitted signals may travel longer distances at higher speeds.
- The shield minimizes electrical and radio frequency interference.
- Coaxial cabling is the primary type of cabling used by the **cable television industry** and is also widely used for **computer networks**, such as Ethernet.

Coaxial Cable



- Coaxial cable has two wires of copper.
- The core/inner copper wire in centre and is made of solid conductor. It is enclosed in an insulating sheath.
- The second/outer copper wire is wrapped around, and is used to protect from external electromagnetic interference(Noise).
- This all is covered by plastic cover used to protect the inner layers from physical damage such as fire or water.

Coaxial Cable Standards

- Coaxial cables are categorized by their Radio Government (RG) ratings. Each RG number denotes a unique set of physical specifications

- | | | |
|----------|---------------|------------------------------|
| • 50-Ohm | RG-7 or RG-11 | : used with thick Ethernet. |
| • 50-Ohm | RG-58 | : used with thin Ethernet |
| • 75-Ohm | RG-59 | : used with cable television |

- | | | |
|----------|-------|------------------------------|
| • 12-Ohm | RG-29 | : used with cable television |
| • 20-Ohm | RG-28 | : used with thin Ethernet |



50 Ohm



75 Ohm



RG CABLE

RG-58 C/U



RG-59 B/U



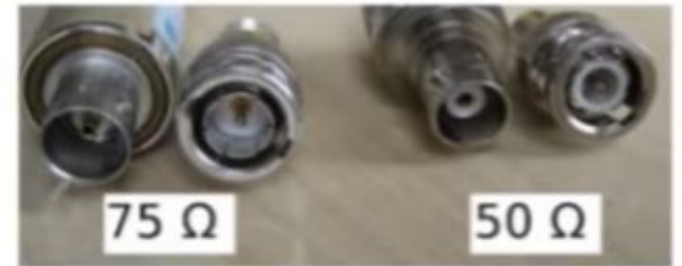
RG-62 A/U



RG-6/U



RG-11/U



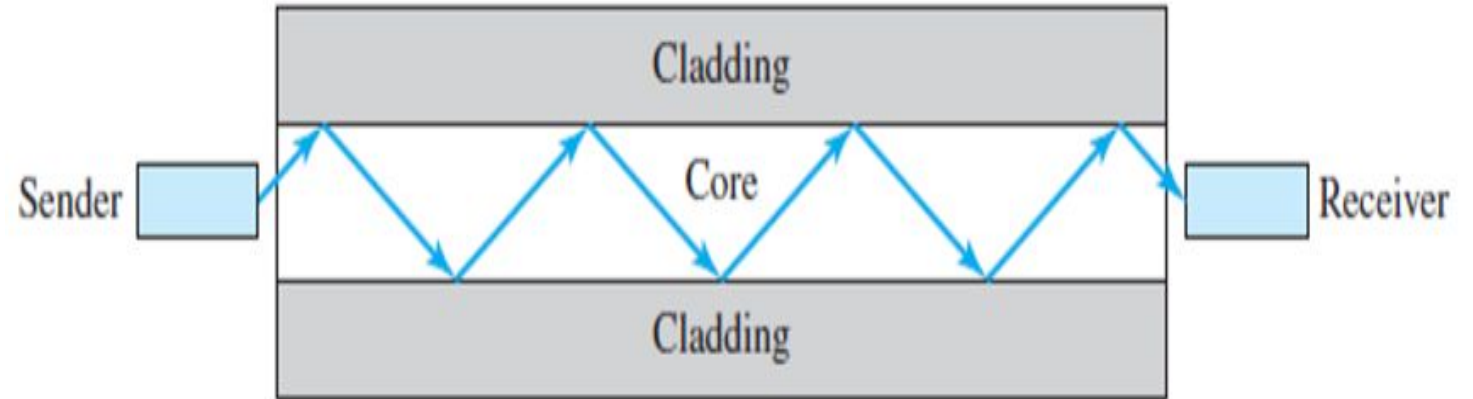
COAXIAL CABLE (contd...)

Applications

- Analog telephone networks
- Cable TV networks
- Traditional Ethernet LAN – 10Base2, 10Base5

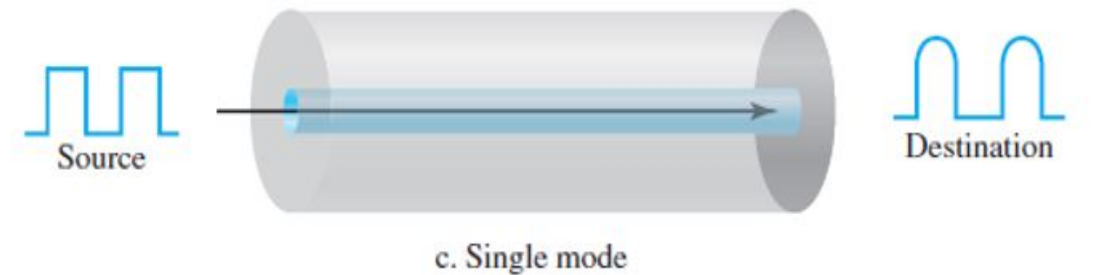
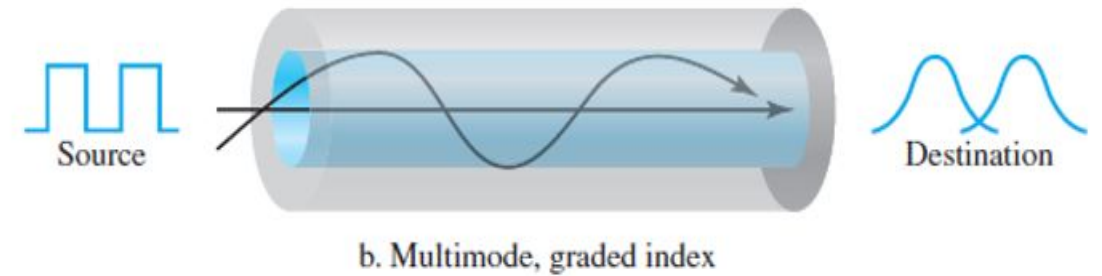
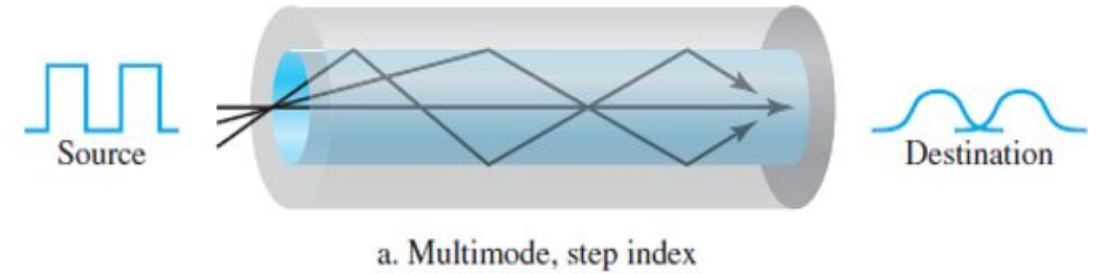
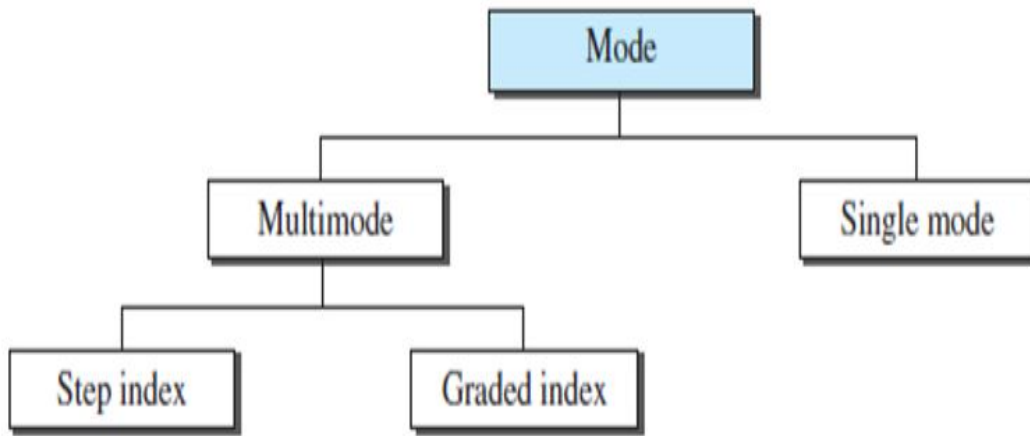
FIBER-OPTIC CABLE

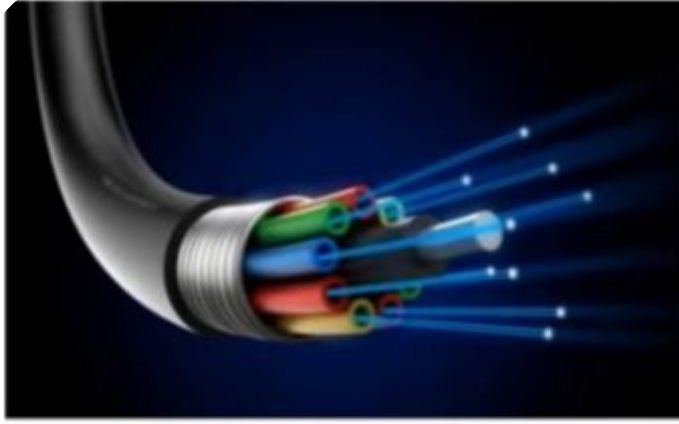
- Higher bandwidth
- Less expensive
- Immune to electrical noise
- More secure – easy to notice an attempt to intercept signal
- Physical characterizes
 - Glass or plastic fibers
 - Very thin (thinner than human hair)
 - Material is light



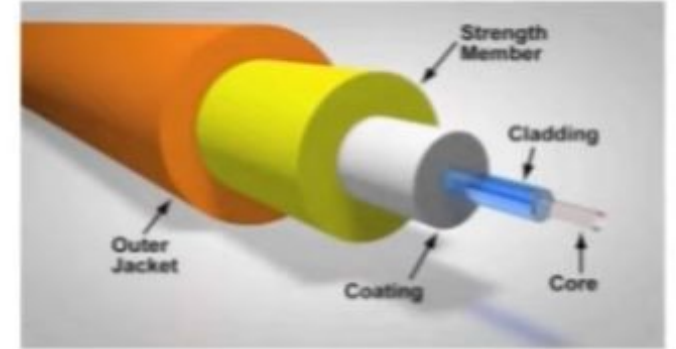
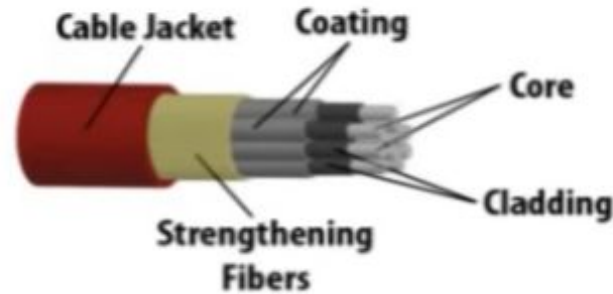
FIBER-OPTIC CABLE (contd...)

Propagation modes





Fibre Optics Cable



- A fibre optic cable is made of high quality of **thin glass or plastic** and is used to transfer digital data signals in the form of **light** up to distance of thousand of miles.
- Fibre optic cables are **not affected** by electromagnetic interference, so noise and distortion is very less.
- Fibre optic cables carry communication signals using pulses of light generated by **small lasers** or **light-emitting diodes** (LEDs).
- The cable consists of one or more strands of glass, each only slightly thicker than a human hair. The centre of each strand is called the **core**, which provides the **pathway for light to travel**. The core is surrounded by a layer of glass called **cladding** that reflects light inward to avoid loss of signal and allow the light to pass through bends in the cable. No light escapes the glass core because of this reflective **cladding**.

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ADVANTAGES AND DISADVANTAGES OF OPTICAL FIBER

Advantages:

- Higher bandwidth
- Less signal attenuation
- Immunity to electromagnetic interference
- Resistance to corrosive materials
- Light weight
- Greater immunity to tapping

ADVANTAGES AND DISADVANTAGES OF OPTICAL FIBER (contd...)

Disadvantages:

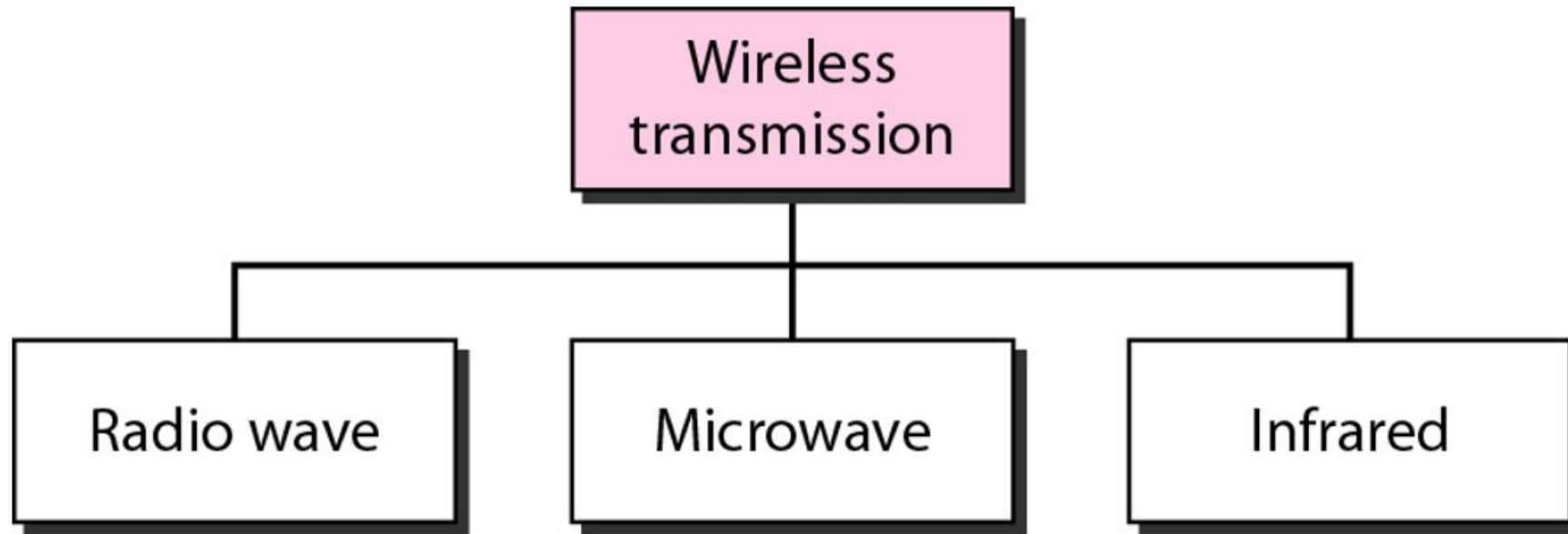
- It is expensive.
- Difficult to install.
- Maintenance is expensive and difficult

FIBER-OPTIC CABLE (contd...)

Applications:

- Backbone networks – SONET
- Cable TV – backbone
- LAN
- 100Base-FX network (Fast Ethernet)
- 1000Base-X

Radio waves



Wireless (Unguided/Unbound) Transmission Media

- A **wave** can be described as a disturbance that travels through a medium from one location to another location.
- A **wave** is a **transfer of energy**, usually through a form of matter called a **medium**.
- There are a special type of wave that can travel without a medium, called **electromagnetic waves** (also called **EM** waves) , which are waves like radio waves and microwaves.
- Unlike sound waves and water waves, electromagnetic waves don't need a fluid, or a solid, or even air to help them travel from one place to another. EM waves can travel across the great vacuum of space, which is why we see light from distant stars and planets.

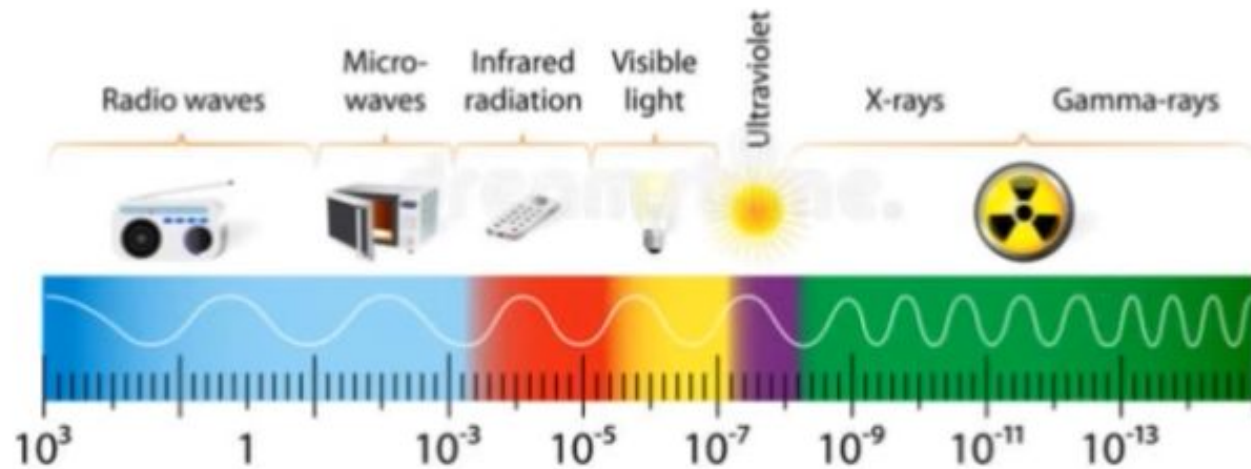
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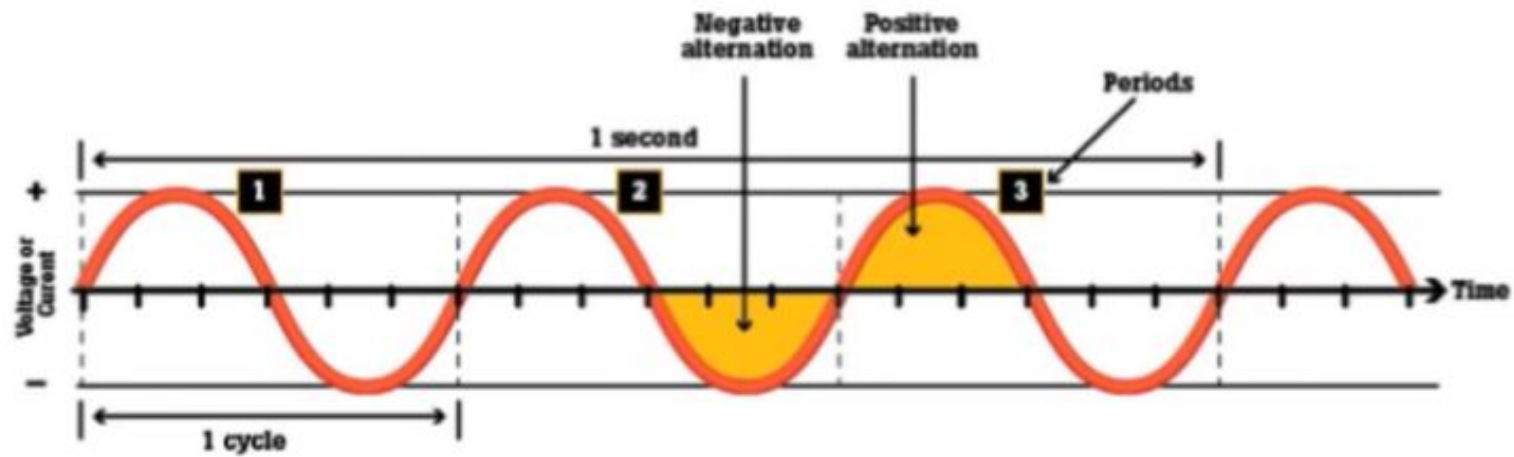
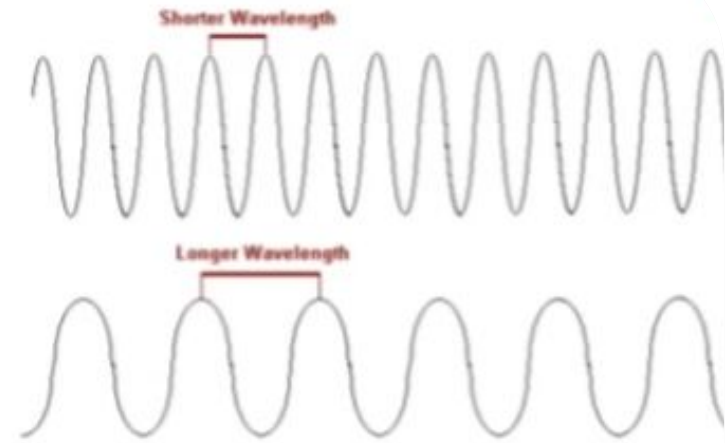
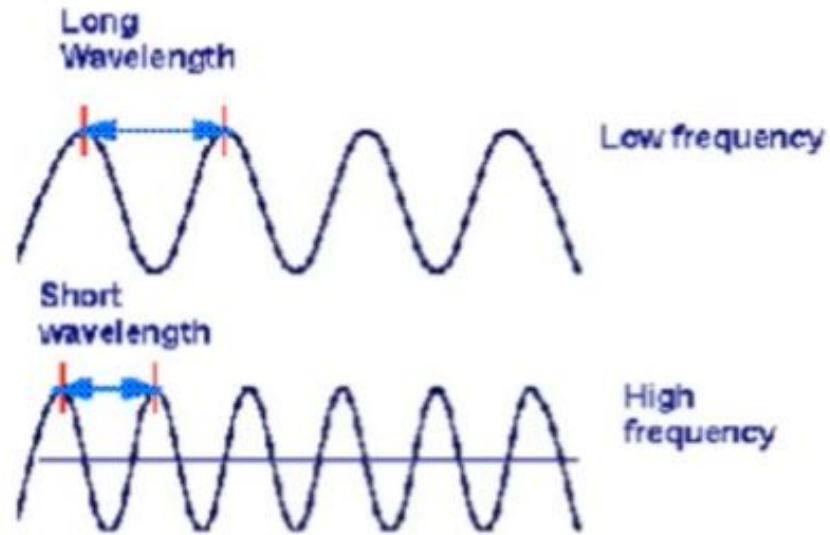
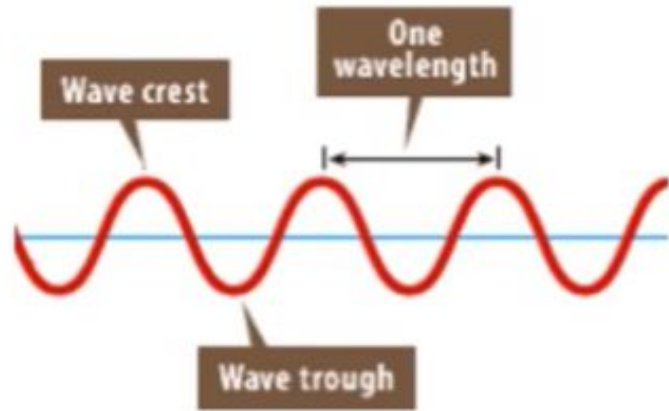
- **Electromagnetic waves** are formed when an **electric field** comes in contact with a **magnetic field**. They are hence known as '**electromagnetic**' waves.
- Electromagnetic (EM) radiation is a form of energy that is all around us and takes many forms, such as radio waves, microwaves, X-rays and gamma rays.
- **Sunlight** is also a form of EM energy. Electromagnetic energy from the sun comes to Earth in the form of radiation.
- **The Electromagnetic Spectrum** describes a wide range of different **electromagnetic waves**.

Wireless (Unguided/Unbound) Transmission Media

- A little part of **electromagnetic spectrum** can be used for wireless transmission.

THE ELECTROMAGNETIC SPECTRUM





Frequency = 3 hz

Period = 3 μs

Band	Frequency range	Wavelength range
Extremely Low Frequency (ELF)	<3 kHz	>100 km
Very Low Frequency (VLF)	3 to 30 kHz	10 to 100 km
Low Frequency (LF)	30 to 300 kHz	1 m to 10 km
Medium Frequency (MF)	300 kHz to 3 MHz	100 m to 1 km
High Frequency (HF)	3 to 30 MHz	10 to 100 m
Very High Frequency (VHF)	30 to 300 MHz	1 to 10 m
Ultra High Frequency (UHF)	300 MHz to 3 GHz	10 cm to 1 m
Super High Frequency (SHF)	3 to 30 GHz	1 to 1 cm
Extremely High Frequency (EHF)	30 to 300 GHz	1 mm to 1 cm

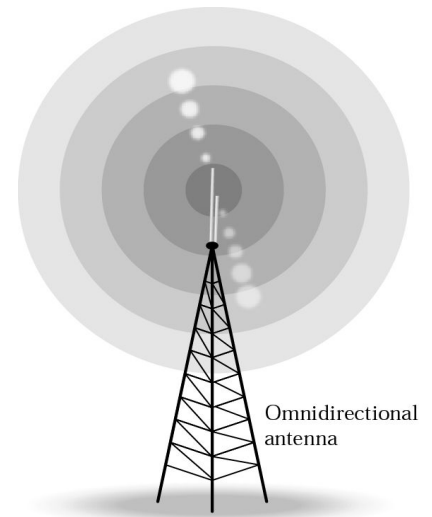
Radio Waves Transmission

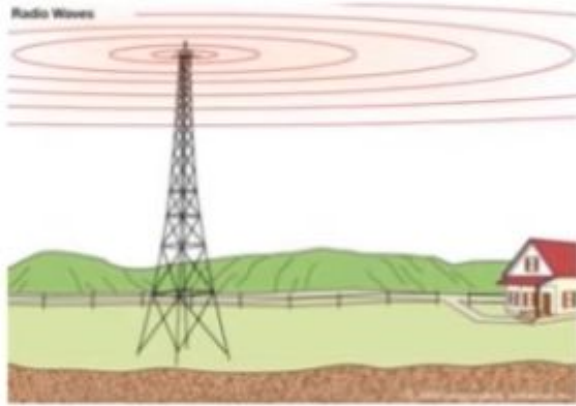


- Radio waves are **EM (Electromagnetic) waves** that have wavelengths between 1 millimetre and 100 kilometres (or 300 GHz and 3 kHz in frequency).
- Radio frequency is easy to generate because it has large wavelength and can travel long distance.
- Radio waves are generated by **radio transmitters** and received by **radio receivers**.
- Radio stations transmit radio waves using **transmitters**, which are received by the **receiver** installed in our devices. Both transmitters and receivers use antennas to radiate or capture radio signals.
- It can penetrate walls easily, so these waves are widely used for communication both indoors and outdoors.

Radio waves

- ☐ **Frequency range- 30 MHz to 1 GHz**
- ☐ **Omnidirectional in nature.**
- ☐ **Can penetrate walls**
- ☐ **Less sensitive to rainfall**
- ☐ **Not Suitable for line of sight communication.**





Radio Waves Transmission



- Radio waves are **omnidirectional** means they travel in all the directions from the source.
- When an antenna transmits radio waves, they are propagated in all directions.
- A sending antenna send waves that can be received by **any receiving antenna**. The omnidirectional property has disadvantage, too. The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signal using the same frequency or band.
- It is Used Mobile, AM/FM radio, television

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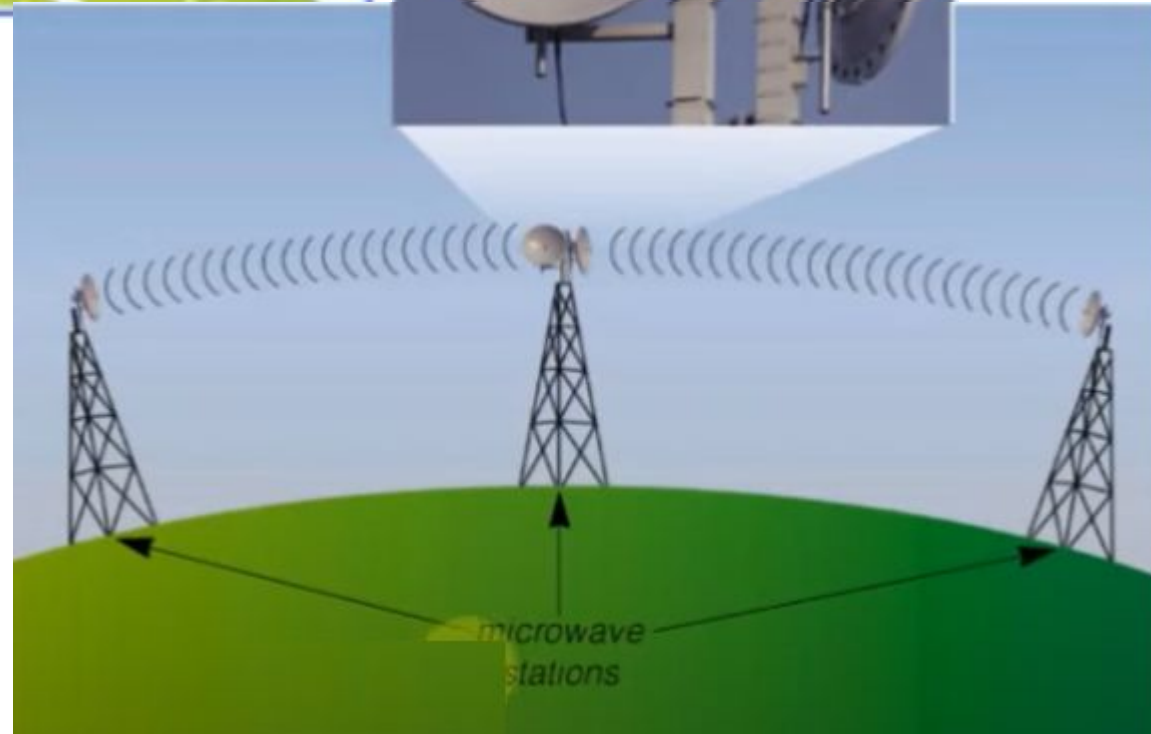
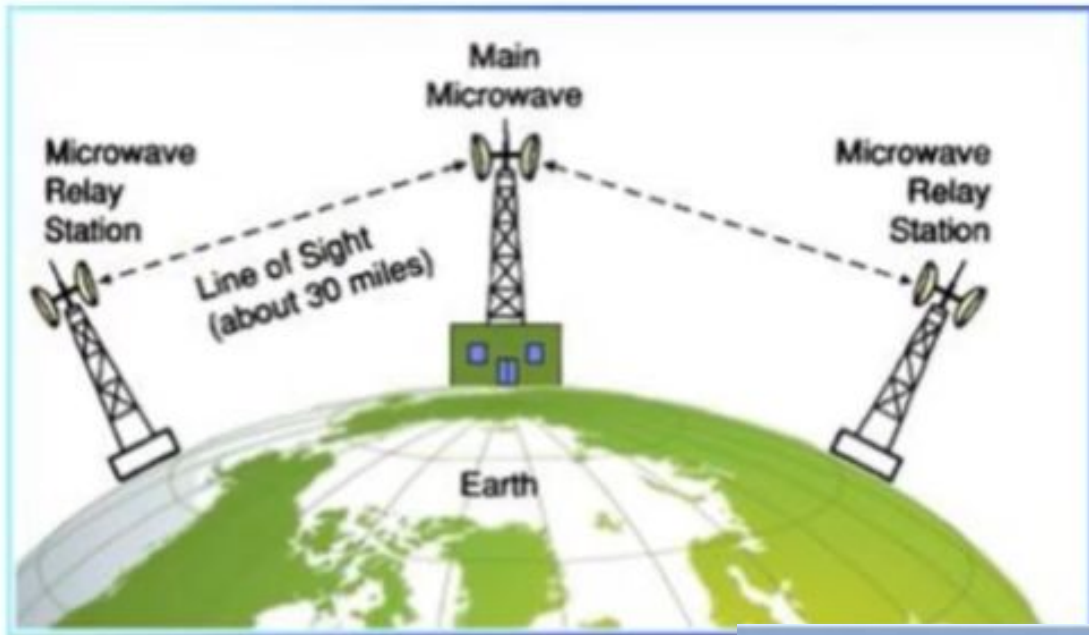
Radio Spectrum

Radio Band	Frequency	Some Applications
Very Low Frequency VLF	3 KHz to 30 KHz	Radio Navigation
Low Frequency LF	30 KHz to 300 KHz	Long Wave Radio
Medium Frequency MF	30 KHz to 3 MHz	AM Radio
High Frequency HF	3 MHz to 30 MHz	CB Radio (HAM) Point to Point Radio Search and Rescue Services
Very High Frequency VHF	30 MHz to 300 MHz	FM radio 88-108 MHz VHF Broadcast TV
Ultra High Frequency UHF	300 MHz to 3 GHz	UHF Broadcast TV Cellular Phones Microwave Links Wi-Fi in 2.4 Band Satellite Communications
Super High Frequency SHF	3 GHz to 30 GHz	Microwave Links Wi-Fi in 5 GHz Band Satellite Communications
Extra High Frequency EHF	30 GHz to 300 GHz	Microwave Links

Micro Waves Transmission

Micro Waves Transmission

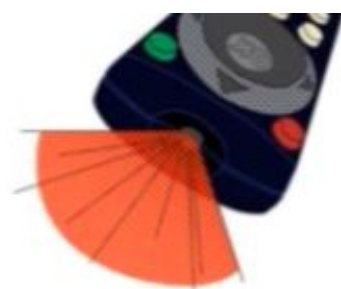
- **Microwaves are a type of radio waves with high frequencies. It can be classified as a subclass of radio waves. The frequency of microwaves lies in the 300 MHz to 300 GHz.**
- **Unlike radio waves, microwaves are unidirectional, in which the sending and receiving antennas need to be aligned.**
- **Microwaves are widely used for point-to-point communications because their small wavelength, which means that the signal is focused into a narrow beam. Additionally, each antenna must be within line of sight of the next antenna.**
- **Electromagnetic waves above 100 MHz tend to travel in a straight line and signals over them can be sent by beaming those waves towards one particular station. Because Microwaves travels in straight lines, both sender and receiver must be aligned to be strictly in line-of-sight.**



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- Microwaves have higher frequencies and do not penetrate wall like obstacles.
- It is used for satellite communication, navigation, radar, remote sensing and other short distance communication systems.

Infrared Waves Transmission



- Infrared signals have frequencies between 300 GHz to 400 THz. They are used for short-range communication.
- Infrared waves are used for very short distance communication like TV remote, wireless speakers, automatic doors, hand held devices etc.
- Infrared waves having high frequencies prevents interference b/w one system to another.
- Infrared signals have high frequencies and cannot penetrate walls. Due to its short-range communication system, the use of an infrared communication system in one room will not be affected by the use of another system in the next room. This is why using an infrared TV remote control in our home will not interfere with the use of our neighbour's infrared TV remote control.
- ***THE DISADVANTAGES OF USING INFRARED***
 - Infrared signals cannot be used for long distance communication. In addition, we cannot use infrared waves outside a building because sun's rays contain infrared waves that can interfere with communication.

Satellite Communication

- ❑ **A communication satellite is basically a microwave relay station.**
- ❑ **Used to link two or more earth stations.**
- ❑ **Satellite communication configurations:**
 1. **Point to point configuration**
 2. **Broadcast configuration**

